

TEJAS NALADALA

MACHINE LEARNING ENGINEER / EMPIRICAL RESEARCHER / HARDWARE FOUNDER

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I am 19 and study Engineering at the University of Washington. I founded R0 Systems, build machine-learning evaluation systems, and run empirical studies in embedding evaluation, agent search, reinforcement learning, medical imaging, and ocean computer vision. I have published three peer-reviewed plasma-engineering papers.

EDUCATION

University of Washington, Seattle

AUG 2025 - JUN 2028 EXPECTED

Engineering

- + Lavin Entrepreneurship Program, a selective multi-year Buerk Center cohort.
- + Coursework and independent work spanning ML, signal processing, embedded systems, optimization, and research methods.

WORK AND VENTURES

Ro Systems

JUN 2024 - PRESENT

Founder and CEO / plasma-activated-water sanitation for post-harvest produce

- + Bootstrapped the company to six figures in revenue.
- + Architected the continuous-flow machine stack: patent-pending plasma reactor, 20 kV electronics, STM32 firmware, venturi gas-liquid transfer, process-water handling, mechanical assemblies, validation, and field deployment.
- + Published three peer-reviewed papers, has two patents pending, and completed independent validation with national food-safety laboratories.
- + Demonstrated up to 3x shelf-life extension at under \$0.50 per ton in energy cost across tomato and leafy-green studies.

OpenTrade (YC S26)

JUN 2026 - SEP 2026

Machine Learning Engineer / first ML hire

- + Built the path from live market data to grounded LLM-generated trade cards and the personalized ranking that determined what each user saw.
- + Built factual-accuracy and regression checks for grounding, numerical consistency, temporal validity, unsupported claims, and structured output.
- + Turned impressions and downstream behavior into evaluation and training signals for a product that reached roughly 150,000 monthly users in two months.

SEAL Lab, University of Washington

MAR 2025 - NOV 2025

Research Associate / PI: Prof. Alex Mamishev

- + Built a PPG drowsiness wearable with 95% motion-artifact rejection and a Navy hull-integrity piezoelectric sensor with real-time 512-point FFT at 10 kHz and acoustic-emission classification.
- + Co-authored technical sections of two funded grant proposals and surveyed 40+ papers on PPG processing and embedded anomaly detection.

CSIR-NIIST, Centre for Sustainable Energy Technologies

JUN 2024 - MAR 2025

Research Associate / PI: Dr. Suraj Soman

- + Fabricated dye-sensitized and perovskite solar cells using FTO substrates, TiO₂ photoanodes, thermal processing, ruthenium sensitizers, electrolytes, counter electrodes, and sealed assemblies.
- + Ran J-V curves, IPCE, photoinduced absorption spectroscopy, and XRD with Scherrer crystallite-size analysis.

Ingenium Naturae

MAY 2023 - JUN 2024

Research Engineer

- + Delivered custom plasma and automation machines for agri-food clients across power electronics, embedded control, fluidics, reactor design, and industrial mechanics.
- + Designed 20 kV discharge drivers, MOSFET switching stages, PID control loops, STM32 firmware, electrode and reactor geometries, Venturi injectors, and production CAD assemblies.
- + Handled specifications, architecture, component selection, fabrication, wiring, firmware, validation, acceptance testing, documentation, and handoff.

RESEARCH**MTEB-Gym**

JUN 2026 - PRESENT

Research collaborator / MTEB team

- + Co-developed MTEB-Gym, a label-free framework for ranking embedding models on datasets without human relevance annotations.
- + Worked on bidirectional judging, validation, uncertainty estimates, and failure handling for label-free rankings.
- + Contributed nine merged upstream changes across validation, uncertainty estimation, caching, deterministic parallelism, and failure handling.

RSNA Knee Abnormality Detection

JUL 2026 - PRESENT

Independent research / in progress

- + Built hierarchical 2.5D multiple-instance learning around ordered MRI slices and study-level aggregation.
- + Added patient/study splits, fold-integrity checks, leakage audits, label-quality analysis, and metric-consistency tests to the experiment path.
- + In progress.

maze-rl-baselines

FEB 2026 - APR 2026

Sole author / reproducible benchmark

- + Benchmarked PPO, DQN, and A2C against simple heuristics and behavior cloning on procedurally generated mazes.
- + The public repository includes experiment code, run records, manifests, and a result verifier.
- + Later runs outgrew the pinned manifest; artifact reconciliation is in progress before final reporting.

Connectome Architecture Benchmark

DEC 2025 - PRESENT

Independent audit and corrected protocol

- + Audited the original 757-row benchmark and found failures in trainable-component control, null-graph density, measured-data provenance, artifact binding, and CartPole evaluation.
- + Preserved and invalidated the original rows so the failure remains inspectable.
- + Implemented and tested CAB v2 with readout-only optimization, digest-bound measured connectomes, and edge/weight-matched nulls. The full corrected run is pending.

agentbreed

MAR 2026 - PRESENT

Sole author / preregistered synthetic pilot

- + Typed genome representation and nine search baselines under matched compute across 700 deterministic evaluations, three domains, and 20 seeds.
- + Multi-component search beat prompt-only search (pooled $d_z = 1.32$, p approximately $1e-14$ across 60 seed-domain pairs). Search operators were statistically indistinguishable after correction.
- + The repository contains the tested optimization library, reproduction path, and preregistration for the pending real-LLM replication.

VISIONS '26 / OCEAN 411

AUG 2026

NSF OOI expedition RR2607 / R/V Roger Revelle

- + Fifteen-day operations cruise with 18 ROV Jason dives at Axial Seamount and Southern Hydrate Ridge, instrument work to 2,900 m, CTD casts, chlorophyll filtering, and oxygen, RAS, and PPS microbial-DNA sampling.
- + Built a provenance-aware multimodal expedition memory over video, sonar, hydrophone, CTD, and navigation streams.
- + Building the computer-vision, acoustic-alignment, calibration, and held-out validation pipeline.

PEER-REVIEWED PUBLICATIONS

Design of systems for plasma activated water for agri-food applications
J. Phys. D: Appl. Phys. 57(49), 493003 / 2024
Misra NN, Naladala T, Alzahrani KJ. DOI: 10.1088/1361-6463/ad77de.

Design of a continuous PAW disinfection system for fresh produce industry
Innov. Food Sci. Emerg. Technol. 97, 103845 / 2024
Misra NN, Naladala T, Alzahrani KJ, Sreelakshmi VP, Negi PS. DOI: 10.1016/j.ifset.2024.103845.

Design and construction of continuous industrial-scale cold plasma equipment
Innov. Food Sci. Emerg. Technol. 97, 103840 / 2024
Misra NN, Sreelakshmi VP, Naladala T, Alzahrani KJ, Negi PS. DOI: 10.1016/j.ifset.2024.103840.
Three peer-reviewed papers / 57 citations / h-index 3 as of Sep 2026.

TECHNICAL REPORTS, PATENTS, AND SYSTEMS

VLM Inference Optimization
Technical report / advised by Prof. Zhuang Liu, Princeton University / Jun-Jul 2026
AlphaEvolve-style search on CharXiv chart QA improved the best single-prompt Qwen3-VL-2B configuration to 63.25% held-out. A separately designed per-question router reached 73.0%, compared with the 29.5% naive baseline. Paired McNemar p = 2.4e-6.

Single-GPU Model Serving
Technical report / advised by Prof. Juncheng Yang, Harvard University / Jun 2026
80-point vLLM matrix on one H100: dense/MoE crossover near concurrency 3, Qwen admitting 42 requests versus GLM's 10 at 49K-token generation, and a 20-point GPQA swing from token budget. An in-situ probe caught PCIe-class bandwidth before publication.

Two R0 patents pending
Plasma reactor and continuous-flow system architecture
Continuous plasma-activated-water system integration and dosing architecture.

RECOGNITION AND PROGRAMS

Carnegie Mellon Venture Challenge / 2nd place, national round, 150+ teams / \$4,500 / Mar 2026
UW Science & Technology Showcase / Grand Prize and Best Pitch / \$2,750 / Jan 2026
Red Bull Basement / 2nd place / 2026
Buerk Center Prototype Grant / \$2,000 / 2026
Founders Inc pitch day presenter / Jul 2026

TECHNICAL STACK

Statistics and methods. Pre-registration; Bradley-Terry ranking; mixed effects; TOST; bootstrap CIs; multiple-comparison correction; MAP-Elites; multiple-instance learning.

ML and serving. PyTorch; Hugging Face; PEFT; DDP/FSDP; bf16 and INT4/INT8; vLLM; Stable Baselines 3; Gymnasium; DICOM; RAG.

MTEB-Gym
Public upstream contributions
Label-free embedding evaluation with synthetic queries, bidirectional pairwise judging, Bradley-Terry ranking, and reliability diagnostics.

Procedural-Maze RL Baselines
Artifact reconciliation in progress
Experiment code, run records, manifests, and a result verifier are public; later runs currently exceed the pinned manifest.

AgentBreed
Preregistered synthetic pilot
Seven hundred evaluations of configuration-space richness and search operators under matched compute.

Connectome Architecture Benchmark
Invalidated pilot / corrected protocol pending
The repository preserves the failed artifact and enforces the corrected CAB v2 protocol.

mimic
Robotics / Mar 2026
Browser-to-robot imitation learning: WebRTC teleoperation into a MuJoCo Franka arm, demo recording, ACT or Diffusion Policy training, ONNX export, a configured 60 Hz control loop, and 108 tests.

engram
Neuromorphic learning / Apr 2026
Rust-core continual-learning runtime using local Hebbian and three-factor STDP updates, persistent memory, an action-veto layer, and PyO3 bindings.

Forge
Agent runtime / Feb 2026
Provider-agnostic multi-agent runtime across eight provider families with tools, memory, cost ceilings, FastAPI/WebSocket observability, Python and TypeScript packages, and 70 tests.

WireML
Foundation-model workbench / Apr 2026
Terminal node graph for CLIP or DINOv2 few-shot training and ONNX export across CUDA, MLX, MPS, ROCm, DirectML, and CPU.

delphi-quant
Verification / Jan-May 2026
Retrospective systematic-strategy verification with transaction costs, walk-forward splits, look-ahead checks, multiple-comparison correction, synthetic reproducibility, and structured rejection logs.

a16z Speedrun S008 / R0 Systems / 2026
Y Combinator F26 and Startup School 2026 / on-stage media interview, 1 of 15 among about 6,000 attendees
Z Fellows / 2026
Lavin Entrepreneurship Program / UW Buerk Center
VISIONS '26 expedition / Aug 2026

Systems and hardware. H100/H200 Slurm; Docker; AWS; FastAPI; WebRTC; STM32 and embedded C; power electronics; real-time control; DSP; KiCad; CAD.

Languages. Python; TypeScript; Rust; C/C++; R; SQL; MATLAB. English, Telugu, Hindi, and conversational Mandarin.